

Matrix Theory: Rank of a Matrix

This study note provides a simplified, professional guide to understanding and calculating the **Rank** of a matrix.

1. What is the Rank?

The **Rank** of a matrix is essentially the number of "useful" or linearly independent rows it contains.

- **Key Principle:** Elementary transformations (row/column swaps, multiplication by a constant, or adding rows) do **not** change the rank of a matrix.
 - **Equivalent Matrices:** Two matrices are "equivalent" if one can be transformed into the other via these operations. Equivalent matrices always have the same rank.
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2. Methods to Find Rank

There are three primary methods used in Engineering Mathematics:

1. **Determinant Method** (also known as the Minor Method).
 2. **Echelon Form Method** (using Row Transformations).
 3. **Normal Form Method** (using both Row and Column Transformations).
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3. The Echelon Form

A matrix is considered to be in **Echelon Form** if it follows these three simple rules:

1. **Zero Rows at the Bottom:** All rows consisting entirely of zeros must be below all non-zero rows.
2. **The "Staircase" of Zeros:** In each successive row, the number of leading zeros must increase.
3. **Leading Unity:** The first non-zero element in every row (the pivot) should ideally be **1** (unity).

Determining Rank from Echelon Form

Once a matrix is in Echelon Form:

Rank $\rho(A)$ = The number of non-zero rows.

4. Practical Example

Let's analyze the Echelon matrix provided in the lecture notes:

$$A = \begin{bmatrix} 1 & 5 & 6 & 7 & 0 & 8 \\ 0 & 1 & 2 & 5 & -1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-by-Step Logic Breakdown:

1. Identify Non-Zero Rows:

- Row 1 is non-zero.
- Row 2 is non-zero.
- Row 3 is non-zero.
- Row 4 is a **zero row** (all elements are 0).

2. Verify Rules:

- The zero row is at the bottom.
- The number of zeros before the first non-zero number increases per row (0 → 1 → 3).
- The first non-zero element in every row is 1.

3. Conclusion:

- There are **3** non-zero rows.
- Rank** $\rho(A) = 3$.

5. Summary Table for GitHub Compatibility

Method	Best Used For...	Transformation Used
Determinant	Small 2×2 or 3×3 matrices	Calculation of Minors
Echelon Form	Systems of Equations	Row Transformations
Normal Form	Finding Rank only	Row & Column Transformations